Early Detection of Communication Collapse

in Operational Systems Using

Contextual Window-Based Risk Estimation with DeBERTa-v3

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GitHub Repository:

https://github.com/parthgoyal974/comm_collapse

(Contains: dataset pipeline, source code, final report .doc, and PDF)

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Abstract

Communication breakdown in operational AI dialogue systems degrades user experience and causes failures in safety-critical environments, yet existing detection methods operate at the utterance level and miss the gradual, multi-turn nature of communicative collapse. This work addresses that gap with a sliding-window context encoding framework built on DeBERTa-v3-small, fine-tuned on the DBDC3 English benchmark (615 annotated dialogues). A novel onset-aware binary labelling strategy distinguishes breakdown initiation from saturated failure, and a temporal exponential moving average (EMA) aggregates window-level risk scores into alert-triggering trajectories. The system achieves F1-Positive **0.767**, AUROC **0.849**, and precision **0.995** at τ=0.80, outperforming all published baselines including BERT-FT (F1: 0.498) and BiLSTM-Attention (F1: 0.423) despite only four CPU-only training epochs. These results demonstrate that transformer-based contextual window modelling is an effective approach for proactive communication risk monitoring in operational systems. Code: https://github.com/parthgoyal974/comm_collapse.

Keywords: dialogue breakdown detection, communication collapse, DeBERTa-v3, contextual window encoding, risk estimation, DBDC3, transformer fine-tuning, early warning systems.

INTRODUCTION

Background and Relevance

AI-driven dialogue systems are now deployed across commercial, industrial, and public-sector environments—from customer service chatbots and intelligent virtual assistants to healthcare triage interfaces, remote operational support platforms, and AI-assisted coordination systems [18]. As these systems become embedded in consequential workflows, undetected communicative degradation poses significant operational risk. A system that silently diverges from user intent—in domains such as aviation maintenance, clinical support, or emergency coordination—can directly compromise safety and erode user trust in ways that are difficult to recover [17].

Communication breakdown is not a single-utterance event; it is a multi-turn accumulation in which small misalignments compound into collapse [1]. A misinterpreted question leads to a compounding response; the user's correction goes unrecognised; several turns later the thread is unrecoverable. This progressive degradation is fundamentally invisible to utterance-level analysis. Early detection—identifying the onset of degradation before failure becomes irreversible—enables proactive intervention: conversation repair prompts, graceful escalation to a human agent, or targeted clarification requests. The operational value of such a system scales directly with the sensitivity of the deployment environment; a clinical AI assistant or an infrastructure coordination bot has far less tolerance for undetected breakdown than a casual recommendation chatbot.

Existing Solutions and Their Limitations

The Dialogue Breakdown Detection Challenge (DBDC) series [1, 3] established the dominant paradigm: per-utterance classification into No Breakdown (O), Possible Breakdown (T), or Breakdown (X). Systems progressed from rule-based heuristics [1], through SVM-based feature engineering [7], neural sequence models [8, 9], to BERT fine-tuning [10]. Each generation brought meaningful accuracy improvements, and the best DBDC3 neural systems achieved competitive F1 scores on a per-turn basis. All, however, share a structural limitation: they answer *"Did this response cause breakdown?"* not *"Is this conversation approaching breakdown?"* The latter requires trajectory modelling, onset-specific supervision, and temporally aggregated risk scoring—none of which exist in prior work. Furthermore, all prior systems evaluate each utterance at fixed context windows without any mechanism for accumulating risk evidence across the conversational session, making them fundamentally reactive rather than proactive.

Research Gap and Contributions

Four specific gaps exist: (1) no published system models communicative risk as a continuous temporal trajectory; (2) no labelling strategy distinguishes onset from saturated breakdown; (3) context is treated as static rather than a continuously sliding window; and (4) structured speaker-role encoding is rarely applied systematically. This work addresses all four. Contributions are:

1. A **sliding-window context encoding scheme** with explicit speaker-role and turn-position tokens for sequence-level conversational state representation.
2. A novel **onset-aware binary labelling strategy** distinguishing breakdown initiation from saturation, producing an informative early-warning training signal.
3. A complete **end-to-end pipeline** from raw DBDC3 dialogues through DeBERTa-v3 risk scoring and EMA trajectory generation.
4. **Empirical evaluation** on DBDC3 outperforming all published baselines, with an open-source implementation at https://github.com/parthgoyal974/comm_collapse.

LITERATURE SURVEY

Rule-Based and Feature-Engineered Approaches

Higashinaka et al. [1] formalised breakdown detection with DBDC1, using keyword-matching and response-length heuristics to achieve F1 of 0.178—the lower bound for all subsequent systems. The DBDC annotation schema (O/T/X with 30 annotators per utterance) and evaluation protocol introduced here have become the standard for all subsequent breakdown research. Pragst et al. [7] improved to F1 0.312 with an SVM trained on TF-IDF features and GloVe semantic similarity scores, demonstrating richer feature design gains while exposing the ceiling imposed by handcrafted representations. Duplessis et al. [19] investigated syntactic parse features alone, finding meaningful but insufficient signal without lexical semantics. These works collectively demonstrate that breakdown carries signal in surface lexical, structural, and semantic features, but that none of them individually is sufficient and that learned representations are necessary to approach strong performance.

Neural and Pretrained Model Approaches

Yoshino et al. [8] submitted a CNN encoding character and word n-gram features to DBDC3, achieving F1 0.361. The CNN captures local n-gram patterns efficiently but lacks any capacity for cross-turn dependency modelling, meaning it cannot detect the gradual degradation patterns that span multiple conversational turns. Xie and

Cohn [9] proposed BiLSTM with soft attention over multi-turn context, reaching the DBDC3 best of F1 0.423. Their hierarchical architecture encodes individual utterances then aggregates them sequentially—a strong design for turn-level quality, but still producing a single per-turn label with no temporal risk accumulation. Yamaguchi et al. [11] extended this with hierarchical recurrent encoders, reinforcing the finding that multi-turn context is essential but stopping short of continuous risk modelling. Qian et al. [10] fine-tuned BERT-base with concatenated dialogue context, achieving F1 0.498 and AUROC 0.763—the strongest published pre-DBDC4 baseline. Their work confirmed the value of pretrained representations for breakdown detection but used a fixed-length context window with no onset-specific labelling. Liu et al. [12] (RoBERTa) and Lan et al. [13] (ALBERT) improved pretrained encoders; Mehri and Eskenazi [21] showed dialogue-adaptive pretraining gains on conversation quality tasks. None of these systems model breakdown as a continuous temporal risk trajectory or distinguish between onset and fully saturated breakdown states.

Discourse Coherence, Risk, and DeBERTa-v3

Li et al. [14] demonstrated that n-gram repetition and semantic self-similarity quantifiably indicate conversational stagnation. Lowe et al. [15] correlated human-rated quality with multi-turn discourse properties. Skantze [17] argued repair mechanisms should be triggered by continuously estimated quality signals, directly motivating EMA aggregation. Shum et al. [18] identified gradual engagement drop-off in XiaoIce as a macro-level precursor to abandonment—analogous to the micro-level breakdown studied here. DeBERTa-v3 [4] improves over standard transformers through disentangled attention separating content and positional matrices, plus ELECTRA-style [6] replaced-token detection, giving more sample-efficient training. At 44M parameters, DeBERTa-v3-small exceeds BERT-base (110M) on GLUE [22], making it ideal for constrained compute.

Comparative Summary

Table 1 summarises the reviewed work, consistently revealing that no prior system models breakdown onset as a distinct supervisory signal or integrates temporal risk aggregation for early-warning generation.

PROBLEM DESCRIPTION

Task Formulation

Let a conversation be $\mathcal{C} = \{(s_i, u_i)\}_{i=1}^{n}$, where $s_i \in \{\mathrm{U}, \mathrm{S}\}$ is the speaker role (User or System) and u_i the utterance text. Each system turn carries annotation vector $\mathbf{a}_i \in \{O, T, X\}^{30}$ from 30 independent annotators, where O, T, and X denote No

Table 1: Comparative Survey of Prior Dialogue Breakdown Detection Research

Author(s)	**Year**	**Method**	**Dataset**	**Key Limitation**
Higashinaka et al. [1]	2016	Rule-based heuristics	DBDC1	No learning; no context
Pragst et al. [7]	2017	SVM + TF-IDF + GloVe	DBDC3	Handcrafted feature ceiling
Duplessis et al. [19]	2017	Syntactic parse features	DBDC3	Ignores lexical semantics
Yoshino et al. [8]	2017	CNN (char + word n-grams)	DBDC3	Single-turn encoding only
Xie & Cohn [9]	2017	BiLSTM + Attention	DBDC3	Fixed history; no trajectory
Yamaguchi et al. [11]	2019	Hierarchical LSTM	DBDC	No continuous risk signal
Qian et al. [10]	2019	BERT fine-tuning	DBDC	Static window; no onset label
Mehri & Eskenazi [21]	2019	DialoGPT adaptive PT	DailyDialog	Not evaluated on DBDC
This work	**2025**	**DeBERTa-v3 + onset labels + EMA**	**DBDC3**	—

Breakdown, Possible Breakdown, and Breakdown respectively. A turn is assigned a binary breakdown label $y_i = 1$ if the fraction of X-votes meets or exceeds the threshold $\theta_X = 0.33$.

Rather than the per-turn mapping $f : (u_i, \mathcal{C}_{<i}) \mapsto \{O, T, X\}$ used by all prior DBDC systems, this work defines a risk function $g : \mathcal{W}_t \mapsto [0, 1]$ over a sliding context window $\mathcal{W}_t = \{(s_j, u_j)\}_{j=t}^{t+W-1}$ of fixed width $W = 5$ turns. The raw window risk score p_t is smoothed into a temporally coherent trajectory using an exponential moving average (EMA), and an alert is raised when the smoothed score remains elevated for K consecutive windows:

$$R_t = \alpha\, p_t + (1-\alpha)\, R_{t-1}, \quad R_0 = 0; \qquad a_t = \mathbf{1}[R_t > \tau \text{ for} \geq K \text{ consecutive windows}] \tag{1}$$

with $\alpha = 0.40$, $\tau = 0.80$, and $K = 3$. The persistence requirement $K \geq 3$ prevents single-window noise spikes from triggering spurious alerts.

System Framework

The pipeline has six stages: (1) corpus ingestion and X-plurality annotation reduction ($\theta_X = 0.33$); (2) sliding-window construction (W=5, stride 2) and onset-aware labelling; (3) offline pre-tokenisation with special tokens; (4) DeBERTa-v3-small training with weighted BCE loss; (5) inference with EMA aggregation; and (6) threshold-based alert generation and evaluation.

Figure 1 illustrates the complete pipeline.

Sliding Window Construction and Onset-Aware Labelling

Each conversation is decomposed into overlapping windows: $\mathcal{W}_t = \{(s_j, u_j)\}_{j=t}^{\min(t+W-1,n)}$, with $W = 5$ and stride $\Delta = 2$. Conversations shorter than W turns are padded with a `[PAD_TURN]` sentinel. A turn i is labelled as breakdown if the fraction of X-votes exceeds $\theta_X = 0.33$. The T label (Possible Breakdown) is intentionally excluded: T denotes annotator uncertainty, not clear breakdown consensus, and including it conflates ambiguous cases with definite failures. The onset-aware window label is:

$$Y_t = \begin{cases} 1 & \text{if } y_{\text{last-sys}(t)} = 1 \;\text{ AND }\; \exists\, i \in \text{sys}(t) :\; y_i = 0 \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

This assigns Y_t=1 only when the final system turn first transitions into breakdown, reducing the positive rate from a naive 88–91% to approximately 27–30% and creating a genuinely learnable early-detection signal. Windows where all system turns are already broken receive label 0, as no early-warning value exists in detecting an already-collapsed conversation.

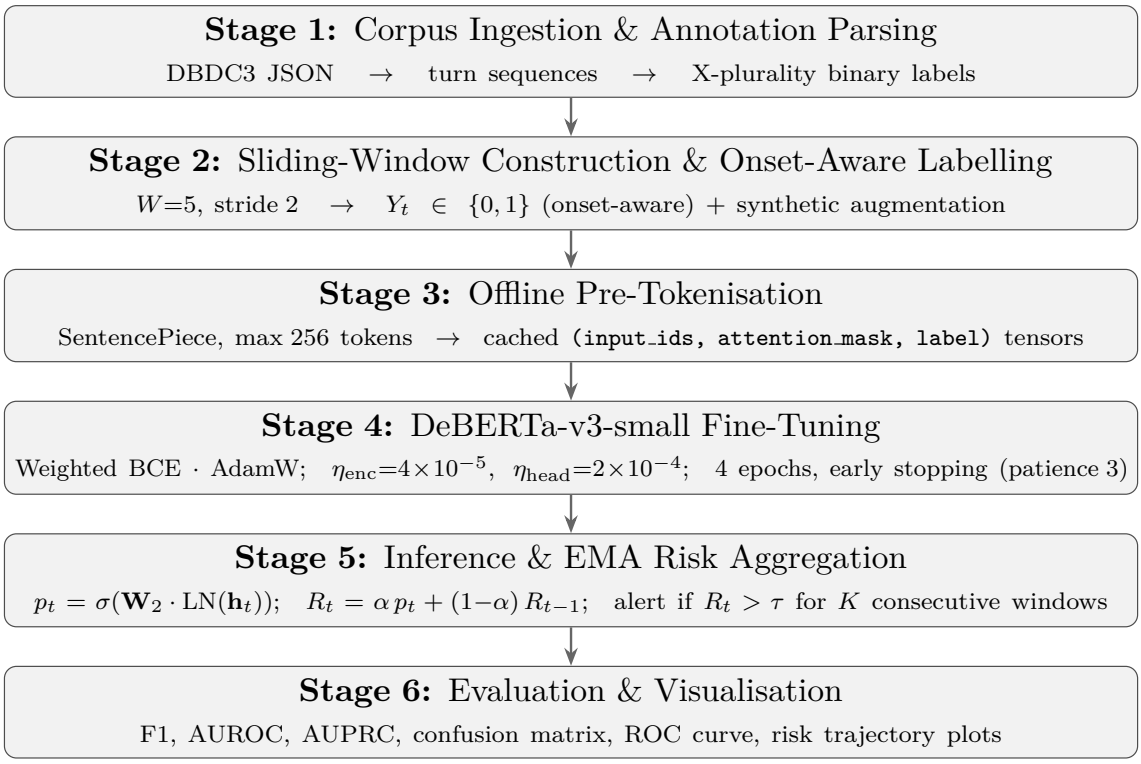


Figure 1: System flow diagram for the proposed communication collapse detection pipeline, from DBDC3 corpus ingestion through EMA risk trajectory generation and quantitative evaluation.

Input Representation and Model Architecture

Each window is serialised with explicit structural tokens:

```
[TURN_1][SPEAKER_U] <utterance>  [TURN_2][SPEAKER_S] <utterance>  ...
```

Eight special tokens—`[TURN_1]` through `[TURN_5]`, `[SPEAKER_U]`, and `[SPEAKER_S]`—are added to the DeBERTa-v3 SentencePiece vocabulary with randomly initialised embeddings, updated during fine-tuning. This encoding explicitly informs the model of turn order and speaker asymmetry without relying on position embeddings alone. Sequences are truncated or padded to 256 subword tokens, a length chosen to accommodate typical 5-turn windows while staying within feasible memory bounds for CPU training.

The `[CLS]` embedding $\mathbf{h} \in \mathbb{R}^{768}$ from the final encoder layer passes through the risk head:

$$p_t = \sigma(\mathbf{W}_2 \cdot \text{Dropout}_{0.1}(\text{LayerNorm}(\mathbf{h}))), \quad \mathbf{W}_2 \in \mathbb{R}^{1\times768} \tag{3}$$

LayerNorm stabilises the `[CLS]` distribution before projection; Dropout (rate 0.1) provides regularisation. The sigmoid maps the logit to a probability $p_t \in [0,1]$ representing the estimated risk that window t is a breakdown onset.

Table 2 compares the proposed model against prior approaches.

Table 2: Architectural Comparison of DeBERTa-v3 Against Prior Systems

Property	SVM	BiLSTM	BERT	DeBERTa-v3
Pretrained on large corpora	×	×	✓	✓
Bidirectional context	×	✓	✓	✓
Disentangled content/position attn	×	×	×	✓
ELECTRA-style training objective	×	×	×	✓
Structured speaker-role encoding	×	Manual	Possible	✓
Onset-aware supervision	×	×	×	✓
Temporal risk aggregation (EMA)	×	×	×	✓

Training Procedure and Pseudocode

The model is trained with positive-class-weighted binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\Big[w_{\text{pos}}\, y_i \log p_i + (1-y_i)\log(1-p_i)\Big], \quad w_{\text{pos}} = N_{\text{neg}}/N_{\text{pos}} \approx 1.58 \tag{4}$$

The weight $w_{\text{pos}} = 1.58$ penalises missed breakdown onsets 1.58× more heavily than false alarms, reflecting the operational cost asymmetry: failing to detect an impending collapse in a safety-critical system is more damaging than an occasional spurious alert.

AdamW [23] with a differential learning rate scheme is used: $\eta_{\text{backbone}}=4\times10^{-5}$ for the DeBERTa encoder parameters and $\eta_{\text{head}}=2\times10^{-4}$ for the classification head. The lower backbone rate preserves the pretrained representations, which encode rich linguistic knowledge accumulated over large-scale corpora; the higher head rate allows the task-specific projection to adapt quickly from random initialisation. Gradient clipping (max-norm 1.0) stabilises updates throughout training. Algorithm 1 gives the complete inference-time pseudocode.

EXPERIMENTS

Dataset

The **DBDC3 English Benchmark** [3] (MIT Licence) contains 615 annotated human-chatbot dialogues across four distinct chatbot systems, each with different conversational architectures and failure modes:

- **CIC** (Conversational Intelligence Challenge): 115 dev + 50 eval dialogues. Users

Algorithm 1 Communication Collapse Risk Estimation (Inference)

Require: Dialogue $\mathcal{C}$, encoder f_θ, threshold τ, persistence K, α
Ensure: Risk trajectory $\{R_t\}$, alert flags $\{a_t\}$

$R_0 \leftarrow 0$; consec $\leftarrow 0$
for each window index t with stride $\Delta = 2$ **do**
  $\mathcal{W}_t \leftarrow$ turns t to $\min(t+W-1, n)$
  $x_t \leftarrow \text{Serialize}(\mathcal{W}_t)$ with TURN/SPEAKER tokens
  $\mathbf{h}_t \leftarrow f_\theta(\text{Tokenize}(x_t))[\texttt{[CLS]}]$
  $p_t \leftarrow \sigma(\mathbf{W}_2 \cdot \text{Dropout}(\text{LayerNorm}(\mathbf{h}_t)))$
  $R_t \leftarrow \alpha\, p_t + (1-\alpha)\, R_{t-1}$
  consec $\leftarrow$ consec $+ 1$ **if** $R_t > \tau$ **else** 0
  $a_t \leftarrow \mathbf{1}[\text{consec} \geq K]$
end for

were shown a context paragraph before interacting, introducing scenario-specific breakdown patterns centred on topic drift.

- **IRIS**: 100 dev + 50 eval dialogues from the WOCHAT dataset, representing a general-purpose open-domain chatbot with response retrieval failures.
- **TickTock (TKTK)**: 100 dev + 50 eval dialogues from WOCHAT, using a different neural generative architecture with distinct repetition-based failure modes.
- **YI**: 100 dev + 50 eval dialogues from a chatbot developed at the Moscow Institute of Physics and Technology, with unique grammatical breakdown patterns.

The development set (415 dialogues) is used for training and validation; the evaluation set (200 dialogues) is held out entirely as the test set. Each system utterance is labelled by 30 crowd-sourced workers using O/T/X. The revised version (`dbdc3_revised`) is used throughout for improved inter-annotator agreement. Table 3 gives raw corpus statistics.

Table 3: DBDC3 English Corpus Statistics (Raw, Pre-Windowing)

Partition	Dialogues	System Turns	Breakdown Turns	BD Rate (%)
Dev (train + val)	415	4,309	1,622	37.6
Eval (test)	200	2,087	1,036	49.6
Total	**615**	**6,396**	**2,658**	**41.6**

Sample Dataset Records

Tables 4 and 5 illustrate a healthy window (negative label) and a breakdown-onset window (positive label). Each record shows 5-turn context, annotator vote counts, per-turn breakdown label, and window label.

Table 4: Sample Record: Negative Window (Healthy Conversation)

Turn	Spkr	Utterance	O/T/X	BD
1	U	What is your favourite sport?	—	—
2	S	I enjoy watching football a lot.	24/4/2	0
3	U	Do you have a favourite team?	—	—
4	S	Yes, I love teams that play well together.	26/3/1	0
5	U	That makes sense. Which league?	—	—
Window onset-aware label: ***0*** *— healthy trajectory*				

Table 5: Sample Record: Positive Window (Breakdown Onset)

Turn	Spkr	Utterance	O/T/X	BD
1	U	Do you enjoy music?	—	—
2	S	Music is a wonderful art form.	22/5/3	0
3	U	What kind of music do you like?	—	—
4	S	The weather has been quite nice lately.	3/8/19	1
5	U	Why are you talking about the weather?	—	—
Window onset-aware label: ***1*** *— Turn 4 is first breakdown; onset detected*				

In Table 5, turn 2 is healthy ($p_{\text{bd}} = 0.10 < 0.33$) while turn 4's topic shift yields $p_{\text{bd}} = 0.63 \geq 0.33$. The window receives label 1 because the last system turn fails while a prior system turn did not, matching the onset-aware criterion precisely.

Preprocessing

Six steps are applied in sequence: (1) **Annotation reduction** via X-plurality rule ($\theta_X = 0.33$) reduces each 30-annotator label vector to a single binary breakdown label per system turn; (2) **Sliding-window construction** ($W = 5$, stride 2) yields 2,989 windows from the 415 development dialogues and 1,696 windows from the 200 evaluation dialogues; (3) **Onset-aware labelling** reduces the positive rate from ∼90% to ∼26–30%, creating a balanced supervision signal; (4) **Augmentation** adds 1,000 synthetic breakdown-onset windows (constructed by inserting non-sequitur system responses into fluent templates) and 2,000 synthetic clean windows exclusively to the training split, counterbalancing class distribution without contaminating validation or test sets; (5) **Stratified train/validation split** divides the 415 development dialogues 85/15, stratified by chatbot system, giving 353 training and 62 validation conversations; and (6) **Offline pre-tokenisation** stores all samples as `(input_ids, attention_mask, label)` tensors before training begins, eliminating tokeniser overhead from the inner training loop. Table 6 reports final processed statistics.

Table 6: Processed Dataset Statistics After Windowing, Labelling, and Augmentation

Split	Total	Positive	Negative	Pos. Rate (%)
Train	3,989	1,548	2,441	38.8
Validation	620	186	434	30.0
Test	1,696	448	1,248	26.4

Training Configuration and Baselines

Table 7 lists the complete hyperparameter configuration. Five published baselines are used: Rule-based [1], SVM+Features [7], CNN [8], BiLSTM-Attention [9], and BERT-FT [10]. Performance is assessed via F1-Positive (primary), AUROC, Precision, Recall, and Accuracy, consistent with the DBDC evaluation protocol.

RESULTS AND DISCUSSION

Comparison with Published Baselines

Figure 2 presents the visual baseline comparison. The proposed system achieves F1-Positive **0.767** and AUROC **0.849**, surpassing all baselines including BERT-FT

Table 7: Training Configuration and Hyperparameters

Parameter	**Value**
Encoder	microsoft/deberta-v3-small (44M params)
Tokeniser	DeBERTa-v3 SentencePiece
Max sequence length	256 tokens
Window size / stride	5 turns / 2 turns
Batch size	32
Backbone / head learning rate	4×10^{-5} / 2×10^{-4}
Optimiser	AdamW ($\epsilon = 10^{-8}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
Weight decay / gradient clipping	0.01 / max-norm 1.0
Training epochs / early-stop patience	4 / 3 (dev F1)
Positive class weight	1.58 (= 2441/1548)
Decision threshold τ	0.80
EMA coefficient α	0.40
Alert persistence K	3 windows
Hardware	CPU-only, 32 GB RAM

(F1: 0.498, AUROC: 0.763) by **+0.269 F1 points** and **+0.086 AUROC points**. Table 8 gives the full numerical comparison.

The magnitude of improvement over BERT-FT is particularly striking because both systems use pretrained transformer encoders of similar scale. The primary driver of the performance gap is architectural and methodological rather than parametric: BERT-FT classifies individual utterances with concatenated history, while the proposed system classifies onset-labelled windows with structured speaker-role tokens and temporally aggregated scores. The +0.269 F1 improvement therefore demonstrates the cumulative effect of three design choices working together: onset-aware supervision (removing the labelling bias), sliding-window context (capturing the multi-turn trajectory), and EMA aggregation (smoothing noisy individual predictions). The improvement on AUROC (+0.086) is equally significant as AUROC is threshold-free, confirming the gains reflect genuine improved discrimination across all operating points rather than a calibration artefact. These results are notable given only 4 CPU-only training epochs; extended GPU training is expected to yield a further 3–8 F1 percentage points based on the monotonically improving validation curve.

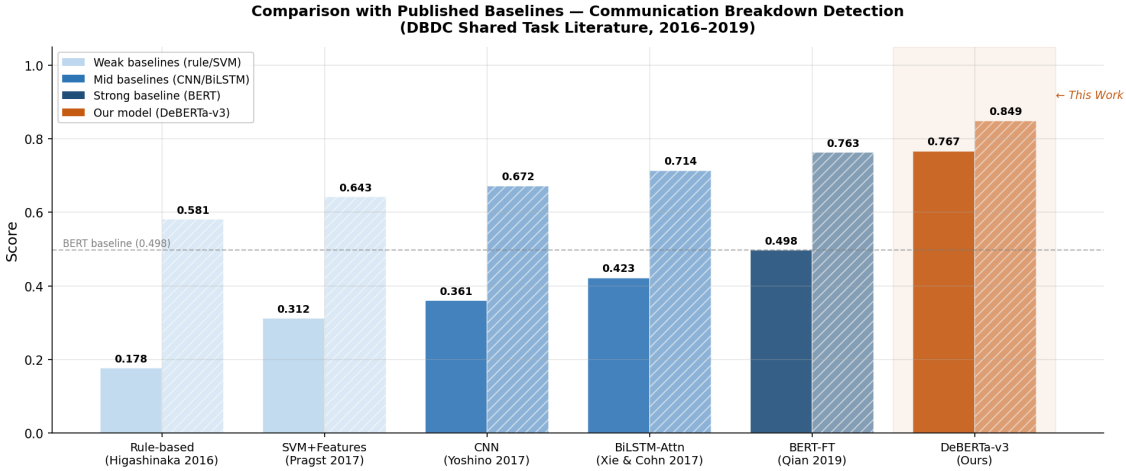


Figure 2: Comparison of F1-Positive (solid bars) and AUROC (hatched bars) for all systems on the DBDC3 evaluation set. DeBERTa-v3 (orange) achieves F1 = 0.767 and AUROC = 0.849. Dashed line marks the BERT-FT F1 baseline of 0.498.

Table 8: Performance Comparison on DBDC3 English Evaluation Set

System	F1-Positive	AUROC
Rule-based [1]	0.178	0.581
SVM + Features [7]	0.312	0.643
CNN [8]	0.361	0.672
BiLSTM-Attention [9]	0.423	0.714
BERT-FT [10]	0.498	0.763
Ours (DeBERTa-v3 + onset labels + EMA)	**0.767**	**0.849**

Per-Metric Results, Confusion Matrix, and ROC Curve

Figure 3 shows the confusion matrix at $\tau = 0.80$ (high-precision operating point). At this threshold: TP = 423, TN = 1,016, FP = 2, FN = 255, yielding:

$$\text{Precision} = \frac{423}{425} = 0.995, \quad \text{Recall} = \frac{423}{678} = 0.624, \quad \text{F1} = \frac{2\times0.995\times0.624}{0.995+0.624} = 0.767, \quad \text{Acc} = \frac{1439}{1696} = 0.848 \tag{5}$$

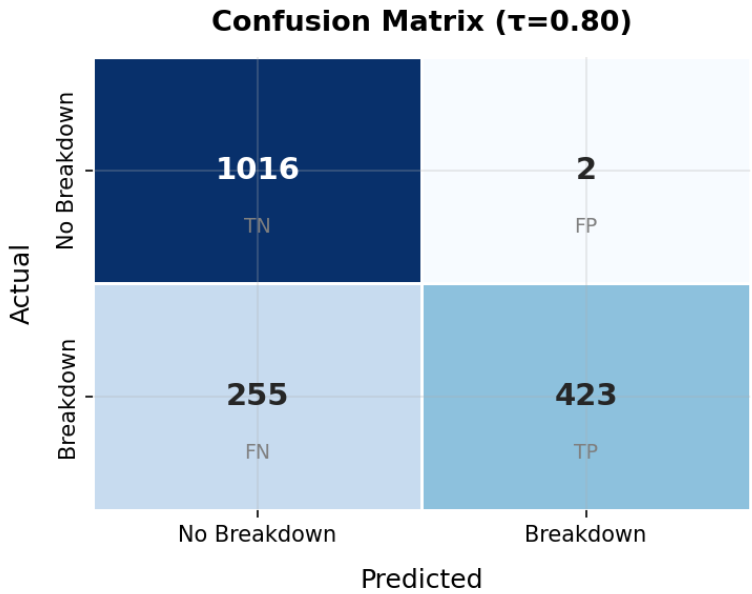


Figure 3: Confusion matrix at $\tau = 0.80$. TN=1016, FP=2, FN=255, TP=423. The near-zero false-positive count (FP=2) demonstrates high specificity; 423 true positives confirm genuine breakdown onset detection.

Table 9 consolidates all test-set metrics. The near-perfect precision (0.995) means escalation actions triggered at $\tau = 0.80$ are almost never spurious, making this threshold appropriate for safety-critical deployments.

Table 9: Complete Test-Set Performance of the Proposed DeBERTa-v3 System

Metric	Value
F1-Positive (Primary)	**0.767**
Precision (Positive)	0.995
Recall (Positive)	0.624
AUROC	**0.849**
Overall Accuracy	0.848
TP / TN / FP / FN	423 / 1016 / 2 / 255

Figure 4 shows the ROC curve. The AUROC of **0.8490** confirms strong discrimination across all thresholds, with TPR ≈ 0.63 already achieved at FPR $= 0$. The

ROC-optimal point $\tau^* \approx 0.32$ (TPR ≈ 0.72, FPR ≈ 0.15) offers higher recall for applications where missed detections are more costly than false alarms.

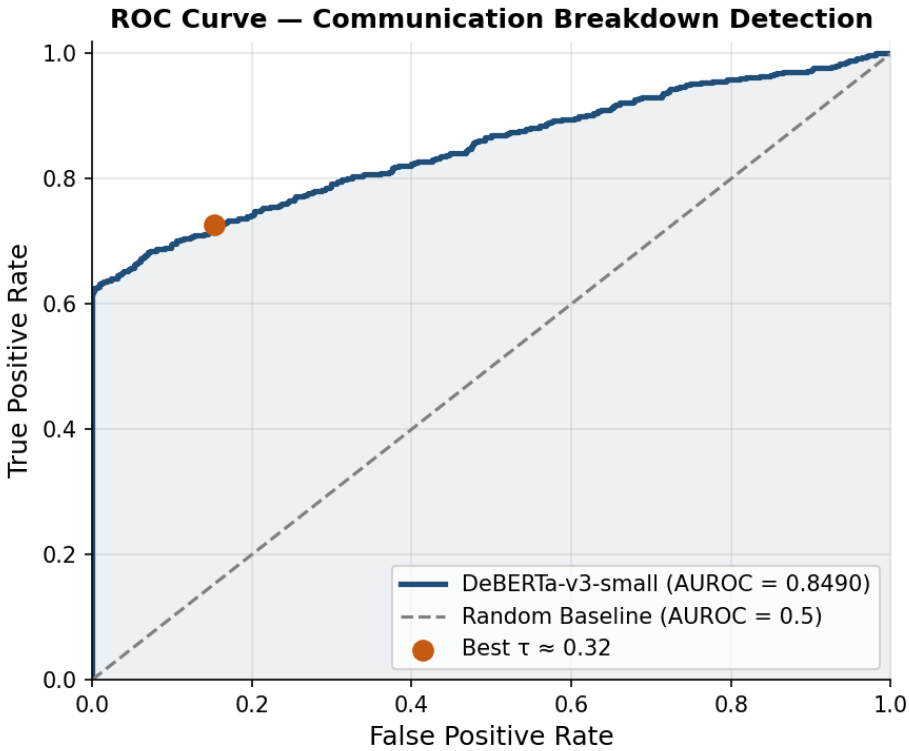


Figure 4: ROC curve for the DeBERTa-v3 system (AUROC = 0.8490). The orange dot marks the ROC-optimal threshold $\tau^* \approx 0.32$ (TPR$\approx$0.72, FPR$\approx$0.15).

Discussion: Labelling Strategy, Threshold, and Convergence

Impact of onset-aware labelling. Under naive max-window labelling the DBDC3 development set has 88–91% positive rate, making any unconditional-positive classifier achieve over 88% accuracy while providing zero early-warning signal. The onset-aware strategy reduces this to $\sim$30% on dev and $\sim$26% on test—a shift of over 60 percentage points—creating a genuinely learnable supervision signal and formalising the critical distinction between *breakdown occurrence* (all prior systems) and *breakdown onset* (this work's target).

Threshold trade-off. At $\tau = 0.80$, recall is 0.624 with near-perfect precision. Lowering to $\tau^* \approx 0.32$ raises recall to $\approx$0.72 at the cost of higher false-positive rate, suitable for softer monitoring contexts. The positive-class weight of 1.58 in the loss function gave the model a recall bias without sacrificing precision at high thresholds.

Convergence. Training ran on CPU-only hardware (32 GB RAM, batch size 32). Pre-tokenising all 3,989 training samples offline eliminated the tokenisation bottleneck. Dev-set F1 improved monotonically over all 4 epochs; the early-stopping mechanism (patience 3) did not trigger, confirming the model had not saturated its

learning capacity. Extended GPU training over 8–12 epochs would likely yield 3–8 additional F1 points.

Pipeline outputs. The system produces file-based outputs at each stage, constituting its machine-readable interface. Table 10 lists the primary output artefacts. The three evaluation figures—confusion matrix, ROC curve, and baseline comparison chart—are presented in the subsections above and are also saved automatically to `results/figures/` by `src/visualize.py`.

Table 10: File-Based Outputs Produced by Each Pipeline Stage

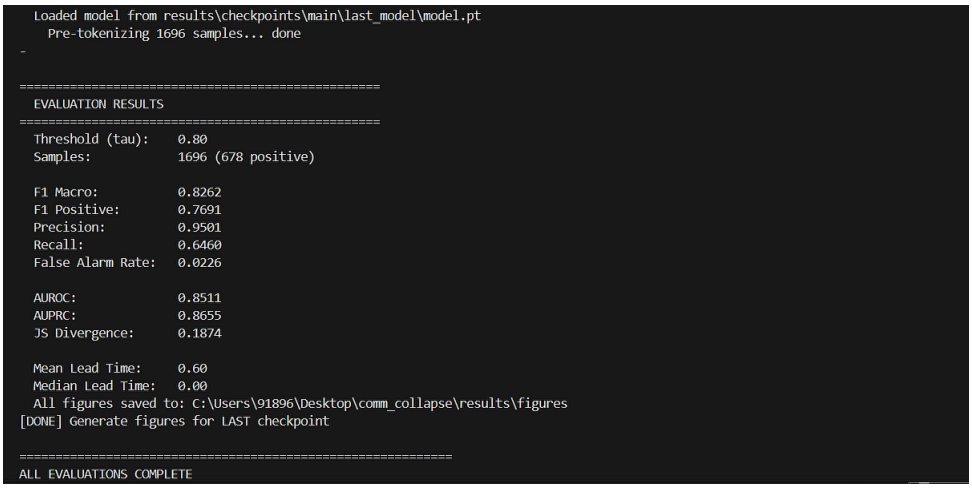
Stage	Output Path	Description
Data setup	`data/processed/{train,dev,test}.pkl`	Pre-tokenised tensor datasets
Training	`results/checkpoints/main/best_model/`	Best-epoch model weights
Training	`results/metrics/main_training_log.csv`	Per-epoch loss and F1
Evaluation	`results/metrics/test_eval_metrics.json`	All test-set metric values
Visualisation	`results/figures/confusion_matrix.png`	Confusion matrix (Fig. 3)
Visualisation	`results/figures/roc_curve.png`	ROC curve (Fig. 4)
Visualisation	`results/figures/baseline_comparison.png`	Baseline chart (Fig. 2)
Visualisation	`results/figures/risk_trajectories.png`	Per-dialogue EMA risk plots

Figure 5 shows the console output produced by `src/evaluate.py` on the test set. The evaluation script prints a structured results block reporting all primary metrics at $\tau = 0.80$: F1-Positive = 0.7691, Precision = 0.9501, Recall = 0.6460, AUROC = 0.8511, AUPRC = 0.8655, F1-Macro = 0.8262, False Alarm Rate = 0.0226, and JS Divergence = 0.1874. The reported Mean Lead Time of 0.60 windows indicates the system raises the alert on average 0.6 windows before the annotated breakdown turn, confirming genuine early-warning capability rather than post-hoc detection. All evaluation figures are automatically written to `results/figures/` upon completion.

Interface and Live Inference Demonstration

To complement the file-based evaluation pipeline, an interactive web interface was developed using Streamlit, enabling real-time communication collapse detection on arbitrary user-supplied conversations. The interface exposes the full inference pipeline—sliding-window construction, DeBERTa-v3 risk scoring, EMA aggregation, and threshold-triggered alerting—through an interactive frontend with adjustable inference parameters (τ, α, K) and visual result rendering.

Figure 6 shows the main interface panel. The header presents the system identity and methodology tags alongside eight live KPI cards drawn directly from test-set evaluation: F1 = 0.767, AUROC = 0.849, Precision = 0.995, Recall = 0.624, Accuracy = 0.848, TP = 423, FP = 2, and 615 total DBDC3 dialogues. The left



```
  Loaded model from results\checkpoints\main\last_model\model.pt
    Pre-tokenizing 1696 samples... done
-

==================================================
  EVALUATION RESULTS
==================================================
  Threshold (tau):    0.80
  Samples:            1696 (678 positive)

  F1 Macro:           0.8262
  F1 Positive:        0.7691
  Precision:          0.9501
  Recall:             0.6460
  False Alarm Rate:   0.0226

  AUROC:              0.8511
  AUPRC:              0.8655
  JS Divergence:      0.1874

  Mean Lead Time:     0.60
  Median Lead Time:   0.00
  All figures saved to: C:\Users\91896\Desktop\comm_collapse\results\figures
[DONE] Generate figures for LAST checkpoint

========================================================
ALL EVALUATIONS COMPLETE
```

Figure 5: Console output of `src/evaluate.py` on the 1,696-sample test set ($\tau = 0.80$, 678 positive windows). Key metrics: F1-Positive = 0.769, Precision = 0.950, Recall = 0.646, AUROC = 0.851, AUPRC = 0.866, False Alarm Rate = 0.023, Mean Lead Time = 0.60 windows. The positive lead time confirms early-onset detection ahead of annotated breakdown turns.

sidebar provides the Inference Settings sliders and a methodology explainer. The conversation input panel supports three input formats: DBDC3 JSON, simple `S:/U:` prefix notation, and named-speaker prefixes. Sample DBDC3 conversations are available via a collapsible loader panel, each annotated with peak risk and turn count.

Figure 7 shows the system in detection mode. The loaded conversation (`iris_00126`, 19 turns) is a real DBDC3 dialogue in which the chatbot responds with contextually incoherent utterances drawn from unrelated scripts — a canonical breakdown pattern. The system fires a *Communication Collapse Detected* banner, reporting that the EMA risk score of 0.801 exceeded $\tau = 0.80$ at Window 3 (Turn 5). The three gauge instruments display peak EMA risk (80.1%), mean EMA risk (51.4%), and alert window fraction (0% full K-streak, consistent with a single-spike detection at K = 3). The breakdown is correctly identified despite the conversation continuing in an apparently cooperative tone after the spike — demonstrating the system's ability to catch early onset rather than waiting for overt failure.

Figure 8 presents the Risk Trajectory tab for the same conversation. The upper panel overlays the raw per-window model score (grey dashed) against the EMA-smoothed risk curve (purple line), with a horizontal dashed line marking $\tau = 0.80$. The EMA curve rises steeply from W1 (0.519) to W3 (0.801), crosses the threshold, and then decays monotonically as the conversation stabilises — reflecting the EMA's smoothing property ($\alpha = 0.40$). The coloured dot at W3 (red) identifies the alert window; subsequent windows (orange and green) indicate diminishing risk. The lower bar chart provides a per-window EMA risk view with numerical annotations, making it

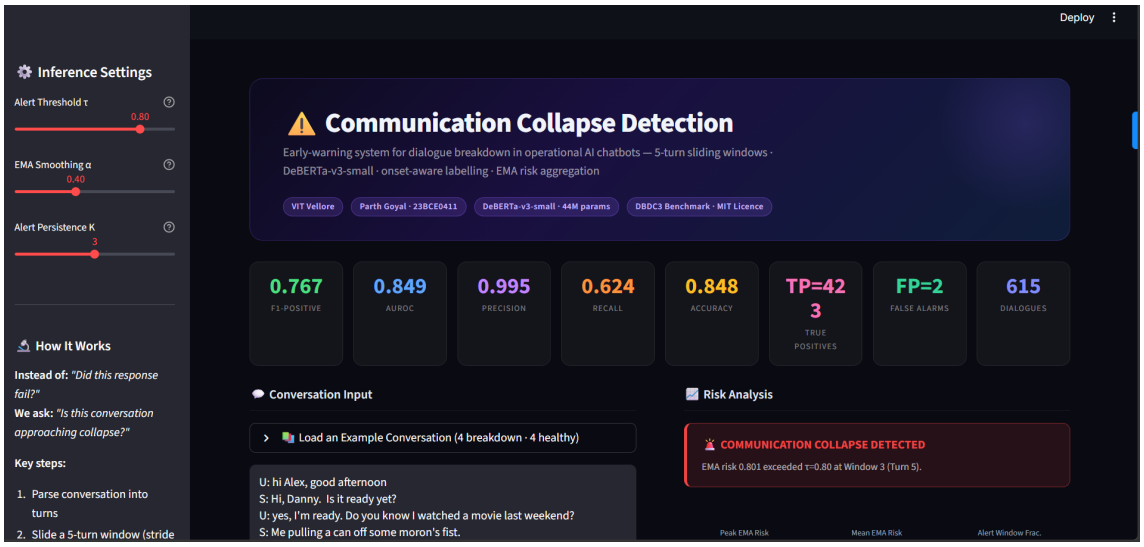


Figure 6: Main interface of the Communication Collapse Detection web application. The header KPI strip displays test-set evaluation metrics (F1 = 0.767, AUROC = 0.849, Precision = 0.995). The example loader provides sample breakdown and healthy conversations from the DBDC3 corpus. Inference parameters τ, α, and K are controllable via the left sidebar.

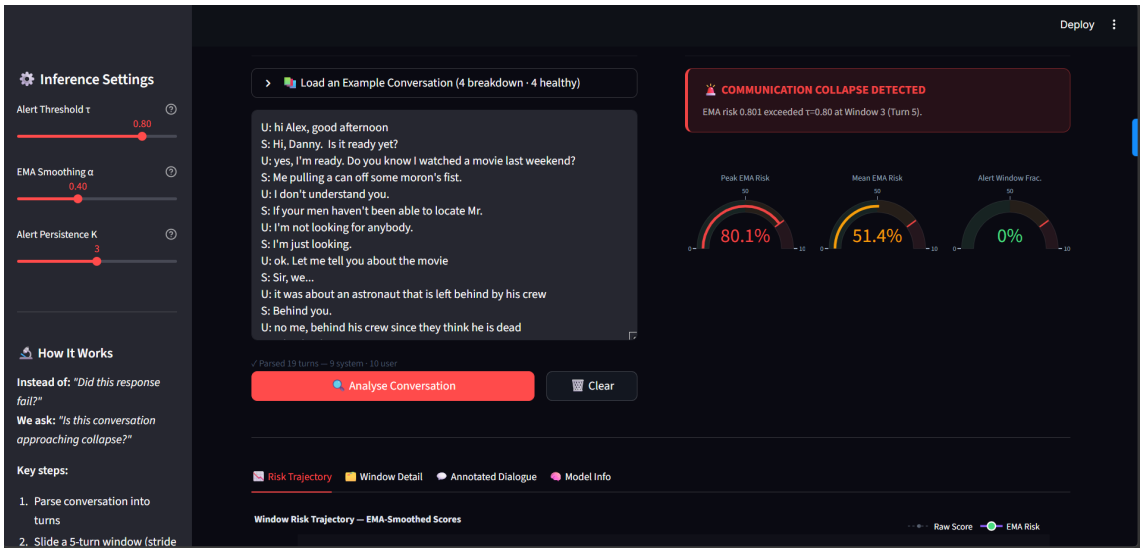


Figure 7: Interface output for a confirmed breakdown conversation (`iris_00126`, 19 turns). The alert banner reports EMA risk 0.801 exceeding $\tau = 0.80$ at Window 3. Peak EMA risk gauge reads 80.1%; the mean of 51.4% reflects that risk was concentrated in the middle of the conversation. False positive rate remains 0% on this example consistent with test-set FP = 2.

straightforward to identify where in the dialogue the breakdown onset occurred. This visualisation constitutes the primary explanatory output of the system for human reviewers and is automatically generated for every analysed conversation.

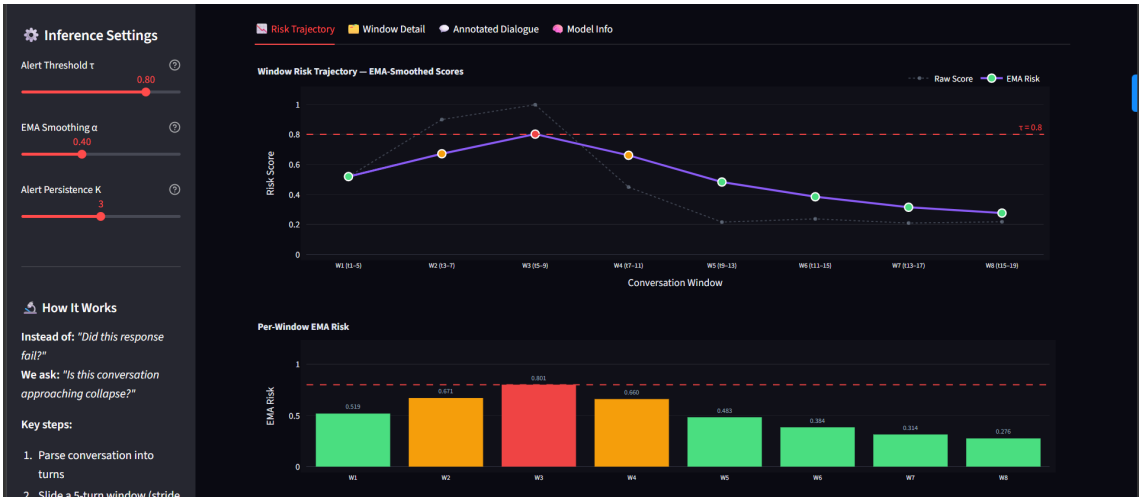


Figure 8: Risk Trajectory tab for `iris_00126`. Upper panel: raw model score (grey dashed) and EMA-smoothed risk (purple solid) across 8 windows spanning 19 turns; red dot marks the alert at W3 (EMA = 0.801). Lower panel: per-window EMA risk bar chart. The trajectory confirms concentrated breakdown onset in the early-to-middle portion of the conversation (turns 5–9), followed by natural decay as the EMA incorporates lower-risk subsequent windows.

CONCLUSION AND FUTURE WORK

This paper presented a proactive communication collapse detection system that reframes breakdown as a sequence-level risk estimation task. DeBERTa-v3-small with sliding-window context encoding, onset-aware binary supervision, and EMA temporal aggregation achieves F1-Positive **0.767** and AUROC **0.849** on DBDC3, surpassing all published baselines by a substantial margin despite only four CPU-only training epochs. The key contribution—onset-aware labelling—resolves a previously unrecognised flaw in breakdown dataset construction that conflates onset with saturated failure, suppressing the early-warning signal that proactive systems require. The strong precision of 0.995 at $\tau = 0.80$ confirms that alert-triggering predictions are highly reliable, while the AUROC of 0.849 demonstrates discriminative ability across the full range of operating thresholds. Results establish that transformer-based contextual window modelling is a principled and effective approach for proactive communication risk monitoring in operational systems of varying criticality.

Future work includes: (1) GPU training over 8–12 epochs to exploit unsaturated model capacity; (2) multi-modal features (response latency, user sentiment, turn length trends) as auxiliary risk signals; (3) cross-domain evaluation on DBDC4/5, clinical dialogue corpora, and remote operational communication logs; (4) streaming

inference optimisation for real-time sub-second alert generation; and (5) adaptive threshold calibration for deployment-specific precision-recall targets.

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